

μFluidic Final Project

Objective: Leverage the high-resolution printing capabilities of resin printers in the Digital Fabrication Lab to produce microfluidic (ufluidic) mixing devices. Groups of 2 will work together to design a chip in Fusion 360. Students will resin print a mold and cast their device with polydimethylsiloxane (PDMS) before sealing it with glass.

Deliverables due Friday (7/19) by 5 pm:

- A uFluidic device that functions as a homogenous mixer. (1 device per 2 group members)
- A 1000-word writeup, which includes pictures of the mold and functioning device. More details are below for the content. (1 writeup per person)

Research: Using the resources below perform background research to determine a suitable geometry for homogenous mixing and to learn more about creating ufluidic devices through 3D printing.

- Relevant papers:
 - [Microfluidic Mixing: A Review. Int J Mol Sci.12\(5\): 3263–3287.](#)
 - Focus on the passive mixing schemes (starting on sect. 3 pg. 3272)
 - [Rapid prototyping of cell culture microdevices using parylene-coated 3D prints Lab Chip, 2021, 21, 4814-4822](#)
 - [PDMS Curing Inhibition on 3D-Printed Molds](#)
 - [PDMS Bonding Techniques for Microfluidics](#)
 - Sections 3.1, 4.1, and 4.2
- Relevant commercial products:
 - [Darwin Microfluidics: Micromixers Chips](#)

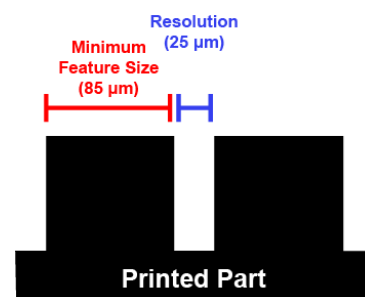
CAD Modeling: A key learning objective of this module is to develop CAD models with dimensions that are informed by the chosen manufacturing process. If the uFluidic channels are modeled too small then they will not be printed. If they are too large, then you will lose valuable reagents as they will be held-up in the device.

Printer Specifications:

- Laser Spot Size (i.e., Minimum Feature Size): 85 μm
- Minimum XY Resolution: 25 μm
- Minimum Z Resolution: 25 μm
- Build volume: (145 × 145 × 185) mm

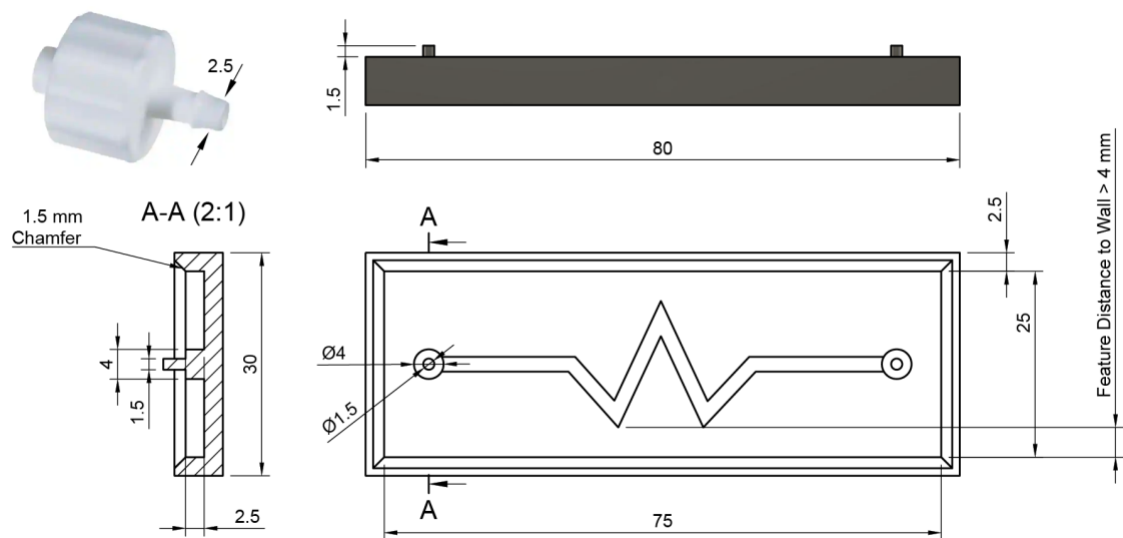
Design Specifications:

- The device should be split up into two sides (one for each team member). Each side must feature at least two different mixing geometries, one of which must have 3D geometry - not just a simple upward extrusion from a sketch (e.g., a zig-zag channel).
 - Work in separate sketches from your team member and label each sketch with your initials. You can only edit the file one at a time.



- **No dead zones.** The fluid flow will take the most direct path to the exit. Channels that double back on themselves or intersect may lead to a different path than expected. Trace your finger through the design, if you skip over a geometry due to a quicker path to the exit, then you have a dead zone.
- The device, when molded, must be precisely 25 mm wide by 75 mm long to sit flush on a [common microscope slide](#) (see dimensioned drawing below).
- Tall and thin components are likely to break off during printing, post-processing, or demolding. **Aspect ratio** is the height to cross-sectional width or length (whichever is smaller) of a printed feature. Keep the aspect ratio less than 4:1. For example a feature that is 1.5 mm in diameter should be no taller than 6 mm.
- For inlets and outlets, use a 1.5 mm post extruded **above the top of the mold**. This will create a hole that is compatible with the microfluidic barb (see image below).
- Channels should be at least 4 mm away from the edge of the device or else leaking will occur.
- The design may draw inspiration from online models, but it must be unique in its combination of mixing elements in its entirety.
- Do not include internal features that cannot be de-molded. This includes cutbacks and small holes (< 0.5 mm).

Important Dimensions for uFluidic Device



Writeup:

- Introduction to uFluidic devices, providing at least one application that requires homogenous mixing. Explain why mixing can be a challenge in uFluidics.
- Describe the logic behind **your** mixing sections.

- How were the dimensions and the shapes of the channels determined? For example, if a zigzag microchannel was selected the width (w) and the step size (s) determine the mixing efficiency.
- How can resin 3D printing be used to make uFluidic devices? What are the benefits and drawbacks of this process?
- Figures
 - Picture of the mold
 - Picture of casted device

Please note: this is a fast-paced module, and it will take time to print the devices. The devices may not be ready until Friday (7/19).

Only one uFluidic device is required per group. However, each student is responsible for their own unique writeup.